Ranking Reversal Theory Under Candidate-Set Shift in GRN Benchmarking

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ABSTRACT

Gene regulatory network (GRN) benchmarks are routinely used to justify scientific claims about method quality, yet the stability of method *ranking* under plausible evaluation-protocol choices is rarely audited. We supply a finite-margin theory of pairwise ranking reversal, a minimum-shift necessary condition, and a magnitude-based reading of the $\Delta = b \cdot g$ decomposition that distinguishes structurally pinned sign-attribution claims from genuinely empirical content. Empirically, across four protocol axes (candidate set, tissue, reference, gene-symbol mapping) on existing single-cell GRN outputs, cluster-bootstrap CIs over method pairs yield reversal rates 16% (candidate, CI 7–27%), 19% (tissue, CI 9–31%), 32% (reference, CI 20–43%); mapping-policy is 0/165. A cluster-restricted permutation MWU shows the calibration-vs-base-rate magnitude ratio is significantly larger in reversal than non-reversal rows ($r = 0.67$, $p < 2 \times 10^{-4}$). AUROC produces 0/40 tissue-shift reversals where AUPR produces 4/45. The practical recommendation: benchmark rank should not be interpreted as method-intrinsic evidence until protocol stability, margin-magnitude regime, *and* metric choice are reported.

CCS CONCEPTS

• **Applied computing** → **Computational genomics**; • **Theory of computation** → *Mathematical optimization.*

KEYWORDS

gene regulatory networks, evaluation bias, ranking stability, benchmark lottery, cluster bootstrap, double-blind

1 INTRODUCTION

Leaderboard rank in GRN inference is routinely used to justify claims about biological plausibility [1, 2], yet the evaluation pipeline involves several protocol choices — candidate-set restriction, reference network, gene-symbol mapping, tissue context — that are rarely reported or controlled [3–6]. The sensitivity of benchmark conclusions to evaluation choices is widely documented in machine learning more broadly: BLEU rank instability under tokenization [11], "benchmark lottery" effects [12], and the need for cross-dataset rank stability [13, 14]. If ranking is unstable under plausible variation, downstream biological decisions flip: which regulators are prioritized, which model is treated as scientifically credible. The field needs ranking-stability theory and diagnostics, not larger metric tables.

Our three contributions: *(1)* a minimum-shift necessary condition yielding a closed-form "safety margin" per pair (§2); *(2)* a strict separation between the algebraic content of the $\Delta = b \cdot g$ decomposition (sign-attribution claims that are *structurally pinned* when base rates are positive) and its empirical magnitude content (§4.2); *(3)* a multi-axis empirical audit with cluster-bootstrap CIs, three null models, margin-magnitude sweep, BH-FDR cell rates, a learned classifier (honest negative result), and a metric-robustness check (§4).

2 THEORY

Let $\mathcal{M}_m(S, \pi, R)$ be a scalar evaluation metric for method m under candidate set S, mapping π, reference R. For methods A, B the *margin* is $\Delta = \mathcal{M}_A - \mathcal{M}_B$, and for settings $1, 2$ a *pairwise ranking reversal* is the event $\Delta_1 \Delta_2 < 0$. Writing $\Delta = b \cdot g$ with $g := \Delta / b$ and $b > 0$, the finite product rule gives the exact decomposition

$$\Delta_2 - \Delta_1 = (b_2 - b_1)\, g_1 + b_2\, (g_2 - g_1). \tag{1}$$

Proposition 1 (Minimum-shift bound). *For a shift family with $|\delta\Delta| \le B$, no pair with safety margin $|\Delta_1| > B$ can reverse. Furthermore, a necessary condition for reversal of a pair with margin $|\Delta_1|$ is $|b_2 - b_1|\,|g_1| + b_2\,|g_2 - g_1| \ge |\Delta_1|$. Hence, if g is shift-stable ($|g_2 - g_1| \le \varepsilon_g$), any reversing base-rate change satisfies $|b_2 - b_1| \ge (|\Delta_1| - b_2\varepsilon_g)/|g_1|$ (informative when $b_2\varepsilon_g < |\Delta_1|$).*

*Structurally pinned sign tallies (*not *findings).* Under $b_1, b_2 > 0$, the "only-b-changes" counterfactual $\Delta_1(b_2 g_1) < 0$ reduces to $(b_2/b_1)\Delta_1^2 < 0$ — *always false*; the "only-g-changes" counterfactual $\Delta_1(b_1 g_2) < 0$ reduces to $(b_1/b_2)\Delta_1\Delta_2 < 0$ — *always true* on reversal rows. Earlier versions of this analysis ([6] and references therein) reported these as findings; they re-state the reversal definition. The genuinely empirical decomposition content is the *magnitude* relationship between the two terms (§4.2).

3 METHODS

Data. Benchmark medians of AUPR/AUROC/F1 across three Tabula Sapiens tissues [7] (kidney, lung, immune) × three candidate-set definitions, on six methods (GENIE3 [5], GRNBOOST2, PEARSON, SPEARMAN, SCGPT_ATTENTION [1], RANDOM); immune-slice reference shifts use HPN_DREAM, BEELINE_GSD [3], and a DOROTHEA [8]/TRRUST [9] union; mapping shifts use four policies of a probe-prior subset (also covering OMNIPATH [10]).

Analyses. (i) pairwise reversal indicator $\mathbf{1}\{\Delta_1\Delta_2 < 0\}$ per (method-pair, shift-cell) per axis; (ii) **cluster-bootstrap 95% CIs** (2,000 resamples; cluster: method pair) [15], replacing Wilson intervals that assume row independence; (iii) magnitude decomposition with **cluster-restricted permutation MWU** (5,000 within-cluster label permutations); (iv) per-cell BH-FDR; (v) per-axis margin-magnitude threshold sweep with cluster-bootstrap bands; (vi) three nulls: joint method relabel, score-noise bootstrap (σ=10%× within-cell median margin), within-cell rank permutation; (vii) LOO-by-tissue instability screening: quantile heuristic vs learned logistic regression on features $|\Delta_1|$, b_1, $b_2 - b_1$, $|g_1|$; (viii) metric robustness over AUPR/AUROC/F1. Seeds fixed (project, perm, bootstrap = 42); decomposition closure verified to 10^{-9}.

Table 1: Headline reversal rates by protocol axis with cluster-bootstrap 95% CIs. Tissue is reported three ways because they answer different questions (see §4.1).

Axis	n	k	Rate	95% CI
Candidate-set shift	135	22	0.16	[0.07, 0.27]
Tissue shift (cand-cond.)	135	26	0.19	[0.09, 0.31]
Tissue shift (dedup, any cand.)	45	17	0.38	[0.18, 0.58]
Tissue shift (dedup, all cand.)	45	1	0.02	[0.00, 0.07]
Reference shift (immune)	106	34	0.32	[0.20, 0.43]
Mapping-policy shift	165	0	0.00	[0.00, 0.00]

4 RESULTS

4.1 Candidate-set and tissue axes

The marginal 16% candidate-shift rate is driven by the immune tissue and small-margin pairs (§4.3). Under per-cell BH-FDR, *none* of the high-reversal cells highlighted in the workshop precursor (e.g. the immune all-pairs to TF-source-target shift at 6/15) survive $q < 0.05$; what survives is the opposite — cells significantly *more stable* than chance (the kidney TF-source to TF-source-target shift at 0/15, $q = 5 \times 10^{-4}$).

Tissue shifts: deduplicating each (method-pair × tissue-pair) observation yields rate 0.38 (any-candidate; 17/45) and 0.02 (all-candidates; 1/45). The 0.19 candidate-conditional rate inflates the signal by counting each pair three times; we report all three for transparency. The dedup-any 0.38 is the honest single-number summary.

4.2 Magnitude decomposition (the only non-pinned empirical content)

In all 22 candidate-shift reversal rows the calibration-term magnitude *exceeds* the base-rate-term magnitude (22/22, rate 1.00), versus 56/113 in non-reversal rows (rate 0.50). Mean magnitude ratio |cal|/|base-rate|: 1.54 (reversal); 0.83 (non-reversal).

A standard iid Mann–Whitney U test gives $p = 3.6 \times 10^{-7}$, but the 135 rows share method pairs across only 15 unique clusters (7 with any reversal), so the iid assumption is invalid. A cluster-restricted permutation MWU (5,000 within-cluster label permutations) gives observed $U = 2075$, rank-biserial effect size $r = 0.67$ (large), and permutation $p < 2 \times 10^{-4}$ (0/5000 null draws exceeded the observed U; null U 95% interval $[1426, 1907]$, observed U above the 97.5% quantile). The cluster-permutation p is the load-bearing significance statement; the iid p is too liberal.

4.3 Margin regime, nulls, screening, and metric choice

Margin sweep. Candidate-shift reversals *vanish* above the 75th-percentile margin (0/34); reference-shift reversals *persist* across margin thresholds (rate in $[0.22, 0.38]$). Candidate-shift instability is therefore a near-tie phenomenon; reference-shift is not.

Three nulls (candidate axis). Joint method-relabel: null mean 0.50, 95% interval $[0.37, 0.61]$; within-cell rank permutation: 0.50, $[0.37, 0.61]$; score-noise (σ = 10% of median margin): noise-inflated rate 0.29, $[0.21, 0.37]$. Observed 0.16 sits far below all three, indicating the rank signal is real and robust to ∼ 10%-of-margin perturbation.

Instability screening (negative). LOO-by-tissue quantile heuristic peaks at F1 = 0.35 (precision 0.24, recall 0.64) with PR-AUC 0.18; a learned logistic regression on $(|\Delta_1|, b_1, b_2{-}b_1, |g_1|)$ gives PR-AUC 0.13, *below* baseline prevalence 0.16. Per-pair instability is not predictable from these features at this data size; population-level reporting is the defensible mode.

Metric robustness. Tissue-shift reversal rates by metric: AUPR 4/45 = 0.09, F1 4/45 = 0.09, AUROC 0/40 = 0.00 (Wilson upper 0.09). AUROC's rank-monotone invariance accounts for its full stability; AUPR's sensitivity to where the operating-rank density lies relative to the positive base rate accounts for its instability. The rank-stability conclusion is metric-conditional.

Worked example (Prop. 1). (a) Kidney; shift from TF-source to TF-source-target candidates; SCGPT_ATTENTION VS SPEARMAN: $|\Delta_1| = 2.85 \times 10^{-2}$ — no shift family with $|\delta\Delta| < 2.85 \times 10^{-2}$ can reverse this pair. *(b)* Immune; shift from all-pairs to TF-source-target; PEARSON VS SPEARMAN: $|\Delta_1| = 8.24 \times 10^{-7}$ — safety margin essentially zero. This is one of the 22 observed reversals, but the 75th-percentile margin cutoff (§4.3) excises pairs like (b); the substantive candidate-shift rate is the post-cutoff 0/34.

Limitations. Medians only (no replicate noise); reference axis uses one tissue; mapping-policy results are slice-limited; Prop. 1 is necessary, not sufficient.

5 DISCUSSION AND RECOMMENDATIONS

Three patterns: *(i)* calibration-magnitude dominance accompanies candidate-axis reversals; *(ii)* candidate-shift instability is a near-tie phenomenon, reference-shift is not; *(iii)* metric choice changes the conclusion (AUROC fully stable, AUPR not). Sign-attribution counts from the $b \cdot g$ decomposition are algebraic consequences of positive base rates and a fixed shift direction [13], not findings.

Recommendations: cluster-bootstrap CIs over method pairs (not Wilson); the margin-magnitude regime alongside any rate; at least two metrics per rank claim (AUPR + AUROC); the calibration-vs-base-rate magnitude ratio.

Reproducibility. https://anonymous.4open.science/r/grn-ranking-reversal-theory-DB8E/

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